SSC MTS 2024 question paper with answers

9 October 2024 · Shift 3

82 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 8 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 16 questions

Q.1

For 8 innings, a cricketer has an average of 68 runs. In the ninth inning, he scores 220 runs. His average score increased by (rounded off to the nearest whole number):

(A) 61 **(B)** 17
(C) 23 **(D)** 85

Q.2

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 9% per year rate of interest for 6 years.

(A) 1660 **(B)** 1580
(C) 1620 **(D)** 1560

Q.3

Find the simple interest (in ₹) on ₹2000 as a sum borrowed at 8% per year rate of interest for 2 years.

(A) 260 **(B)** 320
(C) 280 **(D)** 360

Q.5

A dozen oranges cost ₹96, 2 dozen apples cost ₹240 and 4 pineapples costs ₹180. What is the average cost (in ₹) per piece?

(A) 12.9 **(B)** 12.6
(C) 16.2 **(D)** 19.2

Q.7

A specific amount of money is loaned out at a simple interest rate, resulting in ₹15,000 in 2 years and ₹20,000 in 4 years. The rate percentage per annum is:

(A) 25% **(B)** 20%
(C) 15% **(D)** 18%

Q.8

A shopkeeper bought 10 dozen eggs at the rate of ₹5 per egg. 12 eggs broke in transit. He sold the remaining eggs at the rate of ₹6 per egg. Find his profit percentage.

(A) 15% **(B)** 10%
(C) 8% **(D)** 20%

Q.9

The speed of a boat in still water is 8 km/h, while it takes 2 hours for the boat to cover a distance of 20 km downstream. What is the speed (in km/h) of the stream?

(A) 2
(B) 1.5
(C) 2.5
(D) 1.75

Q.10

A merchant gives 800 g instead of 1000 g to a customer. What is the merchant's percentage profit?

(A) 15%
(B) 20%
(C) 10%
(D) 25%

Q.11

A rectangular sheet of length 28 cm and breadth 16 cm is folded to meet the edges so that it is made into a cylindrical pipe in two possible ways – one along the length and the other along the breadth. The larger volume is how much percentage of smaller volume?

(A) 185%
(B) 175%
(C) 180%
(D) 165%

Q.12

A train is moving in from north to south direction. It overtakes Raj and Madhur who are walking in the same direction at the rate of 2 km/h and 4 km/h in 9 sec and 10 sec, respectively. If the train is x metres long, find the value of x.

(A) 70
(B) 90
(C) 30
(D) 50

Q.13

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 65 kg and 37 kg is:

(A) 51.4
(B) 50.4
(C) 48.4
(D) 49.4

Q.15

A solution of phenyl and water is 28 litres, with phenyl and water in ratio 4 : 3. 21 litres phenyl-water solution which has phenyl to water ratio as 2 : 1 is added to the 28 litres phenyl-water solution. Again 51 litres phenyl-water solution that has phenyl to water ratio as 9 : 8 is added to this. What is ratio of water to phenyl in the final mixture?

(A) 57 : 43
(B) 43 : 57
(C) 41 : 59
(D) 59 : 41

Q.16

The dimensions of a room are 1650 cm × 1350 cm × 900 cm. What should be the size of the longest tape with which the three dimensions can be measured exactly?

(A) 250 cm
(B) 200 cm
(C) 150 cm
(D) 300 cm

Q.17

Each of the two merchants sell an article for ₹1,000. If Merchant A computes his profit on the cost price, while Merchant B computes his profit on the selling price, then they both end up making profits of 25%. By how much is the profit made by Merchant B more than that of Merchant A?

(A) ₹40
(B) ₹45
(C) ₹50
(D) ₹35

Q.18

A conical tent is to accommodate 12 persons. Each person must have 5 m^2 of the space on the ground and 20 m^3 of air to breathe. The height of the tent should be:

(A) 12.5 m
(B) 22 m
(C) 12 m
(D) 13 m

Q.20

The cost of a raw material increased with equal percentage in two years. The price increased to ₹145.20 from ₹120. What is the equal annual percentage increase?

(A) 11.5%
(B) 10.5%
(C) 11%
(D) 10%

Reasoning Ability and Problem Solving 16 questions

Q.1

What should come in place of the question mark (?) in the given series based on the English alphabetical order? CVF, GTJ, KRN, ?, SNV

(A) ROQ
(B) OPS
(C) PRO
(D) OPR

Q.3

What will come in place of the question mark (?) in the following equation if '+' and '–' are interchanged and '×' and '÷' are interchanged? 12 ÷ 2 × 6 – 11 + 12 = ?

(A) 9
(B) 3
(C) 6
(D) 5

Q.4

In a certain code language, 'DEVICE' is coded as 'EDWHDD' and 'DENIAL' is coded as 'EDOHBK'. How will 'DEMAND' be coded in the same language?

(A) EDNZOC
(B) EDNBOC
(C) EDNBMC
(D) EDLZMC

Q.5

7 is related to 33 following a certain logic. Following the same logic, 9 is related to 41. To which of the following is 14 related following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 56
(B) 61
(C) 65
(D) 67

Q.6

What will come in the place of '?' in the following equation, if '×' and '–' are interchanged and '+' and '÷' are interchanged? 16 – 66 + 6 × 26 ÷ 36 = ?

(A) 186
(B) 241
(C) 166
(D) 204

Q.8

In a code language, 'RAM' is written as ‘Q14’ and ‘CAR’ is written as ‘B19’. How will 'MARK' be written in that language?

(A) L30
(B) M29
(C) M30
(D) L26

Q.10

Select the set in which the numbers are related in the same way as are the numbers of the given sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (29, 77) (24, 62)

(A) (19, 47)
(B) (26, 88)
(C) (31, 93)
(D) (22, 86)

Q.11

What should come in place of the question mark (?) in the given series? 1, 2, 5, 12, ?, 58, 121

(A) 20
(B) 15
(C) 27
(D) 25

Q.12

A, B, C, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order) . Only two people sit between F and C when counted from the right of F. Only three people sit between B and G when counted from the right of G. C sits to the immediate right of G. D sits to the immediate right of A. Who sits fourth to the right of E?

(A) G
(B) A
(C) D
(D) F

Q.14

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statement(s). Statements: All sharks are whales. No whale is a dolphin. Conclusions: (I) No shark is a dolphin. (II) No dolphin is a whale.

(A) Only conclusion II follows.
(B) Neither conclusion I nor II follows.
(C) Both conclusions I and II follow.
(D) Only conclusion I follows.

Q.15

A, B, C, D, E, F and G are sitting around a circular table facing away from the centre (but not necessarily in the same order). Only 3 people sit between D and F when counted from the right of D. E and G are both immediate neighbours of A. B sits immediately to the left of C. Only 1 person sits between B and G when counted from the right of G. What is the position of E with respect to B?

(A) Third to the right
(B) Second to the right
(C) Second to the left
(D) Third to the left

Q.16

In a certain code language, ‘GIRL’ is coded as ‘3618’ and ‘ROAD’ is coded as ‘7462’.What is the code for ‘R’ in the given code language?

(A) 1
(B) 8
(C) 3
(D) 6

Q.17
AB10 is related to CD29 in a certain way. In the same way, KL65 is related to MN84. To which of the following is UV25 related, following the same logic?

(A) ZX45
(B) WX45
(C) WX44
(D) ZX44

Q.18
SK –1 is related to OO 3 in a certain way. In the same way, IA 2 is related to EE 6. To which of the following is EO 5 related, following the same logic?

(A) AS –1
(B) AS 1
(C) AS 9
(D) AT 9

Q.19
In a certain code language, 'can you swim' is coded as 'MQ TO LO' and 'He can drive' is coded as 'LO AB PQ'. How is 'can' coded in that code language?

(A) MQ
(B) LO
(C) TO
(D) AB

Q.20
Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some herbs are trees. Some trees are plants. Conclusions: (I) Some herbs are plants. (II) At least some trees are herbs.

(A) Neither conclusion I nor II follows
(B) Only conclusion II follows
(C) Only conclusion I follows
(D) Both conclusion I and II follow

General Awareness 25 questions

Q.1
As of February 2024, who among the following is the Deputy Chairman of Rajya Sabha?

(A) Dharmendra Pradhan
(B) Jagat Prakash Nadda
(C) Harivansh Narayan Singh
(D) Prakash Javadekar

Q.2
Select the INCORRECT statement.

(A) The chromatin material gets organised into rod-shaped chromosomes only when the cell is about to divide.
(B) Chromosomes are composed of DNA and carbohydrates.
(C) DNA molecules contain the information necessary for constructing and organising cells.
(D) Functional segments of DNA are called genes.

Q.3
Which Fundamental Right guarantees protection of interests of the minorities in the matter of their language, script or culture?

(A) Right to Freedom
(B) Right to Equality
(C) Right against Exploitation
(D) Cultural and Educational Rights

Q.4
The Irani Cup is a first-class cricket competition played between the winner of the Ranji Trophy and which other team?

(A) Board President's XI
(B) Rest of India
(C) India B
(D) India A

Q.5
Which of the following is the National waterway no. 4 in India?

(A) Mahanadi and Brahmani rivers and East Coast Canal
(B) Godavari and Krishna rivers along with Kakinada Puducherry
(C) The Brahmaputra river between Sadiya and Dhubri
(D) The Ganga River between Allahabad-Haldia

Q.6
Vinod Mankara's documentary 'Mohinyattathinte Amma' is based on which of the following world renowned Mohiniyattam dancers?

(A) Kalamandalam Kalyanikutty
(B) Sunanda Nair
(C) P Madhuri
(D) Jayaprabha Menon

Q.7
According to Newland's Law of Octaves, every eighth element has properties similar to those of the first element. Which of the following elements has properties similar to those of N?

(A) S
(B) B
(C) Al
(D) P

Q.8
Which state finished second in the first edition of the Khelo India Para Games?

(A) Rajasthan
(B) Uttar Pradesh
(C) Haryana
(D) Madhya Pradesh

Q.9
Who among the following was the founder of Indian Association in Bengal?

(A) Surendra Nath Banerjee
(B) Rash Behari Bose
(C) P Ananda Charlu
(D) Chittaranjan Das

Q.10
Tax, known as bhaga, was levied on which of the following during Mahajanapada period of Ancient India?

(A) Tax on cattle breeders
(B) Tax on merchants
(C) Tax on cultivators
(D) Tax on artisans

Q.11
A 5% decrease in poverty by 2007 was one of main objectives of the _____ plan.

(A) Fourth Five-Year Plan (1966-1974)
(B) Tenth Year Plan (2002-2007)
(C) Ninth Five-Year Plan (1997-2002)
(D) Seventh Five-Year Plan (1985-90)

Q.12
Identify the INCORRECT pair (State and Iron-ore producing region) from the following.

(A) Madhya Pradesh - Ratnagiri
(B) Odisha - Badampahar
(C) Chhattisgarh - Bailadila
(D) Karnataka - Kudremukh

Q.13
Usain Bolt shared the journey of his life in which of the following autobiographies?

(A) Faster Than Lightning
(B) Let Your Mind Run
(C) Race of My Life
(D) Running for My Life

Q.14
The Quwwat-ul-Islam Mosque was constructed in around 1197 AD by the sultan of which of the following dynasties of Delhi Sultanate?

(A) Tughlaq
(B) Lodhi
(C) Mamluk
(D) Khalji

Q.15
Select the odd one from the following saltwater lakes on the basis of their formation?

(A) Chilika lake
(B) Kolleru lake
(C) Sambhar lake
(D) Pulicat lake

Q.16
In 2022, Arjun Singh Dhurve was honoured with the Padma Shri Award for his contribution in which of the following Indian folk-dance forms?

(A) Bihu
(B) Laho
(C) Baiga
(D) Chhau

Q.17
Who among the following was sent by Akbar to lead the Mughal army in the Battle of Haldighati which was fought in 1576?

(A) Raja Man Singh
(B) Raja Maharana Pratap
(C) Raja Medini Rai
(D) Raja Todarmal

Q.18
In October 2023, Indrasena Reddy Nallu has been appointed the Governor of which of the following states in India?

(A) Goa
(B) Kerala
(C) Tripura
(D) Assam

Q.19
Which executive system is present in India?

(A) Collegiate System
(B) Both Parliamentary and Presidential System
(C) Parliamentary System
(D) Presidential System

Q.20
If the tax rate increases with the taxable base, then this type of taxation is called:

(A) progressive
(B) regressive
(C) proportional
(D) digressive

Q.21
Who among the following launched the Home Rule League in September 1916 at Madras?

(A) Annie Besant
(B) Rash Behari Bose
(C) Lokmanya Tilak
(D) Sucheta Kriplani

Q.22

_____ is mainly celebrated by Maharashtrians as New Year's Day.

(A) Gudi Padwa
(B) Bihu
(C) Pongal
(D) Ugadi

Q.23

Krishnarao Shankar Pandit belonged to which of the following music gharanas?

(A) Agra gharana
(B) Mewati gharana
(C) Atrauli gharana
(D) Gwalior gharana

Q.24

According to the Census of India 2011, which states of India have population density more than 1000 persons per sq. km?

(A) Uttar Pradesh and Haryana
(B) Madhya Pradesh and Maharashtra
(C) Bihar and West Bengal
(D) Kerala and Tamil Nadu

Q.25

The microscopic channels, which traverse the cell walls of plant cells and some algal cells, enabling transport and communication between them is called:

(A) desmosomes
(B) lysosomes
(C) plasmodesmata
(D) ribosomes

English Language and Comprehension 25 questions

Q.1

Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select 'No substitution required'. The keys were “lies on the table”.

(A) lay on the table
(B) No substitution required
(C) lie on the table
(D) lying on the table

Q.2

Select the most appropriate ANTONYM of the underlined word in the given sentence. He is habituated to using words spoken with an unusual amount of “venom” when it comes to his in-laws.

(A) benevolence
(B) ambiguousness
(C) satire
(D) irony

Q.3

Select the most appropriate option to fill in the blank. The pencil has a tip made of _____.

(A) lewd
(B) lead
(C) led
(D) steal

Q.4

Select the most appropriate option to fill in the blank. The manager swiftly overtook him, _____ effortlessly past the skinny Englishman, with a practiced ease.

(A) dawdling
(B) sliding
(C) rendering
(D) kindling

Q.5

Select the most appropriate ANTONYM of the given word. Gracious

(A) Keen
(B) Lonely
(C) Rude
(D) Mythical

Q.6

Select the most appropriate verb form to fill in the blank in the given sentence. She _____ her keys on the kitchen counter.

(A) lays
(B) lying
(C) lain
(D) laid

Q.7

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The brave and courageous soldiers / of our motherland / fell into the enemy / and won the battle.

(A) of our motherland
(B) and won the battle
(C) fell into the enemy
(D) The brave and courageous soldiers

Q.8

Select the most appropriate option to fill in the blanks. Whenever addressed, she comes ____ and performs her ____ in a great way.

(A) forth, roll
(B) fourth, role
(C) fourth, roll
(D) forth, role

Q.9

Select the most appropriate meaning of the given idiom. All in the same boat

(A) When everyone is dealing with the same situation
(B) When everybody is travelling in the same vehicle
(C) When everyone has to start all over again
(D) When a task can be accomplished together

Q.10

Select the most appropriate ANTONYM of the given word. jeopardy

(A) mighty
(B) silence
(C) safety
(D) strict

Q.11

Select the INCORRECTLY spelt word.

(A) Celebrate
(B) Outburst
(C) Entertenment
(D) Conscious

Q.12

Select the most appropriate option to fill in the blank. The success and achievements of a school mostly _____ on its experience and resourcefulness.

(A) depend
(B) begin
(C) achieve
(D) credit

Q.13

Select the option that rectifies the spelling error in the following sentence. The poet was able to compose the long poem in a tranquill atmosphere.

(A) trankwil
(B) tranquil
(C) trancuel
(D) tranquel

Q.14

Based on the situation in the given sentence, select the most appropriate idiom for the underlined segment. There are several people who find themselves “a dominating person in a group”

(A) A bone of contention
(B) A white elephant
(C) Cock of the walk
(D) A man of his words

Q.15

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. Manish has being applying / for colleges / ever since he passed / his entrance test in March.

(A) manish has being applying
(B) his entrance test in March
(C) for colleges
(D) ever since he passed

Q.16

Select the most appropriate degree of comparison to fill in the blank. The underground railway network, 'The Tube' in London is said to be _____ transport in London.

(A) more expensive
(B) the expensive
(C) the most expensive
(D) expensive

Q.17

Select the option with the correct spelling to replace the underlined word in the given sentence. Self-esteem is an important "aspekt" of one's self.

(A) aspet
(B) aspect
(C) aspikt
(D) aaspet

Q.18

Select the most appropriate meaning of the underlined phrase. Jared "deliberately set fire to a property" and got arrested by the police.

(A) Arson
(B) Dystrophy
(C) Dementia
(D) Insomnia

Q.19

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don't find any error, mark 'No error' as your answer. He will be / waiting for me when / I reaches his home.

(A) No error
(B) I reaches his home.
(C) He will be
(D) waiting for me when

Q.20

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The new/ smartphone is/absolutely best than/ the latest model.

(A) absolutely best than
(B) The new
(C) the latest model
(D) smartphone is

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. He knew everyone on the island. Whenever his car stopped, half a dozen voices ____ 1 _____ shout out his name. Policemen, agricultural labourers of low social status and priests waved and smiled ____ 2 ____ he passed; everyone greeted him as a brother. They respected his honesty, and belligerence, and above all, they adored his typically Greek scorn and fearlessness when dealing with any _____ 3 ____ of Governmental red tape. On arrival, two of our cases containing linen and other things had been _____ 4 ____ by the Customs on the curious ground that they were merchandise. So, when we moved out to the villa and the problem of bed linen arose, Mother told Spiro _____ 5 _____ our cases in the Customs and asked for his advice.

Select the most appropriate option to fill in blank number 1.

(A) would
(B) should
(C) could
(D) can

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. He knew everyone on the island. Whenever his car stopped, half a dozen voices ____ 1 _____ shout out his name. Policemen, agricultural labourers of low social status and priests waved and smiled ____ 2 ____ he passed; everyone greeted him as a brother. They respected his honesty, and belligerence, and above all, they adored his typically Greek scorn and fearlessness when dealing with any _____ 3 ____ of Governmental red tape. On arrival, two of our cases containing linen and other things had been _____ 4 ____ by the Customs on the curious ground that they were merchandise. So, when we moved out to the villa and the problem of bed linen arose, Mother told Spiro _____ 5 _____ our cases in the Customs and asked for his advice.

Select the most appropriate option to fill in blank number 2.

(A) so
(B) as
(C) thus
(D) nor

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. He knew everyone on the island. Whenever his car stopped, half a dozen voices ____ 1 _____ shout out his name. Policemen, agricultural labourers of low social status and priests waved and smiled ____ 2 ____ he passed; everyone greeted him as a brother. They respected his honesty, and belligerence, and above all, they adored his typically Greek scorn and fearlessness when dealing with any _____ 3 ____ of Governmental red tape. On arrival, two of our cases containing linen and other things had been _____ 4 ____ by the Customs on the curious ground that they were merchandise. So, when we moved out to the villa and the problem of bed linen arose, Mother told Spiro _____ 5 _____ our cases in the Customs and asked for his advice.

Select the most appropriate option to fill in blank number 3.

(A) forum
(B) from
(C) farm
(D) form

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. He knew everyone on the island. Whenever his car stopped, half a dozen voices ____ 1 _____ shout out his name. Policemen, agricultural labourers of low social status and priests waved and smiled ____ 2 ____ he passed; everyone greeted him as a brother. They respected his honesty, and belligerence, and above all, they adored his typically Greek scorn and fearlessness when dealing with any _____ 3 ____ of Governmental red tape. On arrival, two of our cases containing linen and other things had been _____ 4 ____ by the Customs on the curious ground that they were merchandise. So, when we moved out to the villa and the problem of bed linen arose, Mother told Spiro _____ 5 _____ our cases in the Customs and asked for his advice.

Select the most appropriate option to fill in blank number 4.

(A) concealed
(B) contributed
(C) collected
(D) confiscated

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. He knew everyone on the island. Whenever his car stopped, half a dozen voices ____ 1 _____ shout out his name. Policemen, agricultural labourers of low social status and priests waved and smiled ____ 2 ____ he passed; everyone greeted him as a brother. They respected his honesty, and belligerence, and above all, they adored his typically Greek scorn and fearlessness when dealing with any _____ 3 ____ of Governmental red tape. On arrival, two of our cases containing linen and other things had been _____ 4 ____ by the Customs on the curious ground that they were merchandise. So, when we moved out to the villa and the problem of bed linen arose, Mother told Spiro _____ 5 _____ our cases in the Customs and asked for his advice.

Select the most appropriate option to fill in blank number 5.

(A) about

(B) from

(C) behind

(D) within

Answer key

Numerical and Mathematical Ability

Q.1 **(B)**	Q.2 **(C)**	Q.3 **(B)**	Q.5 **(A)**	Q.7 **(A)**	Q.8 **(C)**
Q.9 **(A)**	Q.10 **(D)**	Q.11 **(B)**	Q.12 **(D)**	Q.13 **(D)**	Q.15 **(B)**
Q.16 **(C)**	Q.17 **(C)**	Q.18 **(C)**	Q.20 **(D)**		

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.3 **(B)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(A)**	Q.8 **(A)**
Q.10 **(A)**	Q.11 **(C)**	Q.12 **(C)**	Q.14 **(C)**	Q.15 **(A)**	Q.16 **(D)**
Q.17 **(C)**	Q.18 **(C)**	Q.19 **(B)**	Q.20 **(B)**		

General Awareness

Q.1 **(C)**	Q.2 **(B)**	Q.3 **(D)**	Q.4 **(B)**	Q.5 **(B)**	Q.6 **(A)**
Q.7 **(D)**	Q.8 **(B)**	Q.9 **(A)**	Q.10 **(C)**	Q.11 **(B)**	Q.12 **(A)**
Q.13 **(A)**	Q.14 **(C)**	Q.15 **(C)**	Q.16 **(C)**	Q.17 **(A)**	Q.18 **(C)**
Q.19 **(C)**	Q.20 **(A)**	Q.21 **(A)**	Q.22 **(A)**	Q.23 **(D)**	Q.24 **(C)**
Q.25 **(C)**					

English Language and Comprehension

Q.1 **(D)**	Q.2 **(A)**	Q.3 **(B)**	Q.4 **(B)**	Q.5 **(C)**	Q.6 **(D)**
Q.7 **(C)**	Q.8 **(D)**	Q.9 **(A)**	Q.10 **(C)**	Q.11 **(C)**	Q.12 **(A)**
Q.13 **(B)**	Q.14 **(C)**	Q.15 **(A)**	Q.16 **(C)**	Q.17 **(B)**	Q.18 **(A)**
Q.19 **(B)**	Q.20 **(A)**	Q.21 **(A)**	Q.22 **(B)**	Q.23 **(D)**	Q.24 **(D)**
Q.25 **(A)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (B) 17

The total rises from 8 × 68 = 544 to 544 + 220 = 764, an average of 764/9 ≈ 84.89. The increase is 84.89 − 68 ≈ 17.

Q.2 · Answer (C) 1620

SI = 3000 × 9 × 6/100 = ₹1620.

Q.3 · Answer (B) 320

SI = 2000 × 8 × 2/100 = ₹320.

Q.5 · Answer (A) 12.9

12 oranges, 24 apples and 4 pineapples (40 pieces) cost 96 + 240 + 180 = ₹516, so the average is 516/40 = ₹12.9 a piece.

Q.7 · Answer (A) 25%

The interest for years 3 and 4 is 20,000 − 15,000 = ₹5,000, or ₹2,500 a year. So the principal is 15,000 − 5,000 = ₹10,000 and the rate is 2,500/10,000 = 25%.

Q.8 · Answer (C) 8%

Cost: 120 × ₹5 = ₹600. The remaining 108 eggs sell for 108 × ₹6 = ₹648, a profit of ₹48: 48/600 = 8%.

Q.9 · Answer (A) 2

Downstream speed = 20/2 = 10 km/h, so the stream's speed is 10 − 8 = 2 km/h.

Q.10 · Answer (D) 25%

He gains 200 g on every 800 g he actually gives: 200/800 = 25%.

Q.11 · Answer (B) 175%

If side a becomes the circumference and side b the height, $r = a/(2\pi)$ and $V = \pi r^2 b = a^2 b/(4\pi)$. The volumes are in the ratio $28^2 \times 16 : 16^2 \times 28 = 28 : 16$, so the larger is 175% of the smaller.

Q.12 · Answer (D) 50

With the train's speed v km/h, its length is the relative distance covered: (v − 2) × 5/18 × 9 = (v − 4) × 5/18 × 10, so 9v − 18 = 10v − 40 and v = 22. Then x = 20 × 5/18 × 9 = 50.

Q.13 · Answer (D) 49.4

(40 + 49 + 56 + 65 + 37)/5 = 247/5 = 49.4 kg.

Q.15 · Answer (B) 43 : 57

Phenyl and water in the three lots are 16 and 12, 14 and 7, and 27 and 24 litres, so the final mixture has 57 litres of phenyl and 43 of water: water : phenyl = 43 : 57.

Q.16 · Answer (C) 150 cm

The longest such tape is the HCF: $1650 = 2 \times 3 \times 5^2 \times 11$, $1350 = 2 \times 3^3 \times 5^2$ and $900 = 2^2 \times 3^2 \times 5^2$, so the HCF is $2 \times 3 \times 5^2 = 150$ cm.

Q.17 · Answer (C) ₹50

A: 25% on cost means cost = 1,000/1.25 = ₹800 and a profit of ₹200. B: 25% of the selling price is ₹250. B makes ₹50 more.

Q.18 · Answer (C) 12 m

The base must be 12 × 5 = 60 m^2 and the volume 12 × 20 = 240 m^3. (1/3) × 60 × h = 240, so h = 12 m.

Q.20 · Answer (D) 10%

145.20/120 = 1.21 = 1.1^2, so the price rose 10% each year.

Reasoning Ability and Problem Solving

Q.1 · Answer (D) OPR
First letters move +4 (C, G, K, O, S), second letters –2 (V, T, R, P, N) and third letters +4 (F, J, N, R, V): OPR.

Q.3 · Answer (B) 3
After interchanging the signs: 12 × 2 ÷ 6 + 11 – 12 = 4 + 11 – 12 = 3.

Q.4 · Answer (A) EDNZOC
The letters move +1 and –1 in turn: DEVICE → EDWHDD and DENIAL → EDOHBK. So DEMAND → E D N Z O C (A – 1 wraps round to Z): EDNZOC.

Q.5 · Answer (B) 61
The second number is 4 times the first plus 5: 4 × 7 + 5 = 33 and 4 × 9 + 5 = 41. So 4 × 14 + 5 = 61.

Q.6 · Answer (A) 186
After interchanging the signs: 16 × 66 ÷ 6 – 26 + 36 = 176 – 26 + 36 = 186.

Q.8 · Answer (A) L30
The first letter moves back one and the number is the sum of the other letters' positions: RAM → Q, 1 + 13 = 14; CAR → B, 1 + 18 = 19. So MARK → L, 1 + 18 + 11 = 30: L30.

Q.10 · Answer (A) (19, 47)
The second number is 3 times the first minus 10: 3 × 29 – 10 = 77 and 3 × 24 – 10 = 62. Likewise 3 × 19 – 10 = 47.

Q.11 · Answer (C) 27
Each term is twice the one before plus 0, 1, 2, 3, 4, 5 in turn: 12 × 2 + 3 = 27, then 27 × 2 + 4 = 58 and 58 × 2 + 5 = 121.

Q.12 · Answer (C) D
Number the seats by places to the right of G. C is 1 (immediately right of G), B is 4 (three people between) and F is 5 (C is third to F's right). A and D then take 2 and 3, with D immediately right of A, so E is 6. Fourth to the right of E is seat 3: D.

Q.14 · Answer (C) Both conclusions I and II follow.
All sharks are whales and no whale is a dolphin, so no shark is a dolphin (I). 'No whale is a dolphin' also means no dolphin is a whale (II). Both follow.

Q.15 · Answer (A) Third to the right
Number the seats by places to the right of G. B is 2 (one person between) and C, immediately right of B, is 3. A sits next to G but not at 1, where its other neighbour would be B instead of E; so A is 6 and E is 5. D and F take 1 and 4 with F fourth to D's right: D is 4 and F is 1. E (5) is third to the right of B (2).

Q.16 · Answer (D) 6
R is the only letter in both words and 6 the only digit in both codes, so R = 6.

Q.17 · Answer (C) WX44
Both letters move 2 places forward and the number rises by 19: AB10 → CD29 and KL65 → MN84. So UV25 → WX44.

Q.18 · Answer (C) AS 9
The first letter moves back 4, the second forward 4 and the number rises by 4: SK –1 → OO 3 and IA 2 → EE 6. So EO 5 → AS 9.

Q.19 · Answer (B) LO
'can' is the only word in both sentences and 'LO' the only code in both, so can = LO.

Q.20 · Answer (B) Only conclusion II follows
Herbs and plants are linked only through trees, by two 'some' statements, so I need not be true. 'Some herbs are trees' means some trees are herbs, so II follows.

General Awareness

Q.1 · Answer (C) Harivansh Narayan Singh

Harivansh Narayan Singh has been Deputy Chairman of the Rajya Sabha since 2018, re-elected in 2020.

Q.2 · Answer (B) Chromosomes are composed of DNA and carbohydrates.

Chromosomes are made of DNA and proteins, not carbohydrates; the other three statements are correct.

Q.3 · Answer (D) Cultural and Educational Rights

Article 29, one of the Cultural and Educational Rights, protects any section of citizens with a distinct language, script or culture of its own.

Q.4 · Answer (B) Rest of India

The Irani Cup is played between the Ranji Trophy champions and a Rest of India team.

Q.5 · Answer (B) Godavari and Krishna rivers along with Kakinada Puducherry

National Waterway 4 covers the Godavari and Krishna rivers with the Kakinada-Puducherry canal stretch. NW-1 is the Ganga from Allahabad to Haldia, NW-2 the Brahmaputra from Sadiya to Dhubri and NW-5 the East Coast Canal with the Brahmani and Mahanadi.

Q.6 · Answer (A) Kalamandalam Kalyanikutty

The title means 'Mohiniyattam's mother': Kalamandalam Kalyanikutty Amma, who did much to revive the dance, is known as the mother of Mohiniyattam.

Q.7 · Answer (D) P

Counting N as the first, the eighth element in Newlands' arrangement is P (N, O, F, Na, Mg, Al, Si, P); both are in group 15 today.

Q.8 · Answer (B) Uttar Pradesh

Haryana topped the first Khelo India Para Games (New Delhi, December 2023), with Uttar Pradesh second and Tamil Nadu third.

Q.9 · Answer (A) Surendra Nath Banerjee

Surendranath Banerjee founded the Indian Association in Calcutta in 1876, with Ananda Mohan Bose.

Q.10 · Answer (C) Tax on cultivators

In the mahajanapadas the tax on crops, usually a sixth of the produce, was called bhaga, a share.

Q.11 · Answer (B) Tenth Year Plan (2002-2007)

The Tenth Plan (2002-2007) aimed to cut the poverty ratio by 5 percentage points by 2007, and by 15 points by 2012.

Q.12 · Answer (A) Madhya Pradesh - Ratnagiri

Ratnagiri is in Maharashtra, not Madhya Pradesh. Badampahar (Odisha), Bailadila (Chhattisgarh) and Kudremukh (Karnataka) are iron-ore regions of those states.

Q.13 · Answer (A) Faster Than Lightning

Usain Bolt's autobiography 'Faster than Lightning' was published in 2013.

Q.14 · Answer (C) Mamluk

Qutb-ud-din Aibak, founder of the Mamluk (Slave) dynasty, built the Quwwat-ul-Islam mosque in Delhi in the 1190s.

Q.15 · Answer (C) Sambhar lake

NCERT lists Chilika, Pulicat and Kolleru as lagoons of the coastal areas; Sambhar in Rajasthan is a salt lake in a region of inland drainage, so it formed differently.

Q.16 · Answer (C) Baiga

Arjun Singh Dhurve of Madhya Pradesh received the Padma Shri in 2022 for the Baiga tribal dance.

Q.17 · Answer (A) Raja Man Singh
Raja Man Singh of Amber commanded Akbar's army against Maharana Pratap at Haldighati in 1576.

Q.18 · Answer (C) Tripura
Indrasena Reddy Nallu became Governor of Tripura in October 2023.

Q.19 · Answer (C) Parliamentary System
India has a parliamentary system: the President is the nominal head, and the Council of Ministers, led by the Prime Minister, is responsible to the Lok Sabha.

Q.20 · Answer (A) progressive
A progressive tax charges a higher rate as the taxable base grows; a proportional tax keeps one rate and a regressive tax's rate falls.

Q.21 · Answer (A) Annie Besant
Annie Besant launched her Home Rule League at Madras in September 1916; Tilak had started his at Belgaum in April 1916.

Q.22 · Answer (A) Gudi Padwa
Gudi Padwa, on the first day of Chaitra, is the Marathi New Year; Ugadi is the New Year of Karnataka, Andhra Pradesh and Telangana.

Q.23 · Answer (D) Gwalior gharana
Krishnarao Shankar Pandit was a leading vocalist of the Gwalior gharana.

Q.24 · Answer (C) Bihar and West Bengal
By the 2011 Census only two states had more than 1,000 persons per sq. km: Bihar (1,106) and West Bengal (1,028).

Q.25 · Answer (C) plasmodesmata
Plasmodesmata are fine channels through plant cell walls that link the cytoplasm of neighbouring cells; desmosomes are junctions between animal cells.

English Language and Comprehension

Q.1 · Answer (D) lying on the table
'Were' needs the '-ing' form of 'lie' (to rest): 'The keys were lying on the table.'

Q.2 · Answer (A) benevolence
'Venom' here is spite or malice; its opposite is 'benevolence', goodwill.

Q.3 · Answer (B) lead
A pencil's tip is its 'lead' (in fact graphite); 'led', the past of the verb 'lead', only sounds the same.

Q.4 · Answer (B) sliding
Overtaking swiftly and 'effortlessly' fits 'sliding' past him; 'dawdling' means moving slowly.

Q.5 · Answer (C) Rude
'Gracious' means courteous and kind; its opposite is 'rude'.

Q.6 · Answer (D) laid
SSC's key gives 'laid', the past tense of 'lay' (to put down), which takes an object: 'She laid her keys on the kitchen counter.' 'Lays' would also be grammatical for a habit; 'lying' and 'lain', forms of 'lie', would need a helping verb and take no object.

Q.7 · Answer (C) fell into the enemy
To attack the enemy is to 'fall upon' (or 'fall on') it; 'fall into' means to drop into something or into a state.

Q.8 · Answer (D) forth, role
To 'come forth' is to come forward, and the part one plays is a 'role'; 'fourth' and 'roll' only sound the same.

Q.9 · Answer (A) When everyone is dealing with the same situation
To be 'all in the same boat' is for everyone to be facing the same situation, usually a difficult one.

Q.10 · Answer (C) safety
'Jeopardy' means danger; its opposite is 'safety'.

Q.11 · Answer (C) Entertenment
The correct spelling is 'entertainment'.

Q.12 · Answer (A) depend
Success 'depends on' something; the plural subject 'success and achievements' takes 'depend'.

Q.13 · Answer (B) tranquil
The correct spelling is 'tranquil'.

Q.14 · Answer (C) Cock of the walk
The 'cock of the walk' is someone who dominates a group.

Q.15 · Answer (A) manish has being applying
The present perfect continuous is 'has been applying', not 'has being applying'.

Q.16 · Answer (C) the most expensive
Compared with all the transport in London, the Tube needs the superlative: 'the most expensive'.

Q.17 · Answer (B) aspect
The correct spelling is 'aspect'.

Q.18 · Answer (A) Arson
Arson is the crime of deliberately setting fire to property. Dystrophy is the wasting away of a body part, dementia a loss of memory and reasoning, and insomnia an inability to sleep.

Q.19 · Answer (B) I reaches his home.
The subject 'I' takes 'reach', not 'reaches': 'when I reach his home'.

Q.20 · Answer (A) absolutely best than
'Than' needs the comparative: 'better than', not 'best than'.

Q.21 · Answer (A) would
'Would' describes something that happened again and again in the past: whenever his car stopped, voices 'would shout out' his name.

Q.22 · Answer (B) as
They waved and smiled 'as' he passed, that is at the moment he passed.

Q.23 · Answer (D) form
'Any form of Governmental red tape' means any kind of it; 'forum', 'from' and 'farm' only look alike.

Q.24 · Answer (D) confiscated
The Customs had 'confiscated', that is seized, the cases, on the ground that they were merchandise.

Q.25 · Answer (A) about
One tells someone 'about' something: Mother told Spiro about the cases held in the Customs.

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